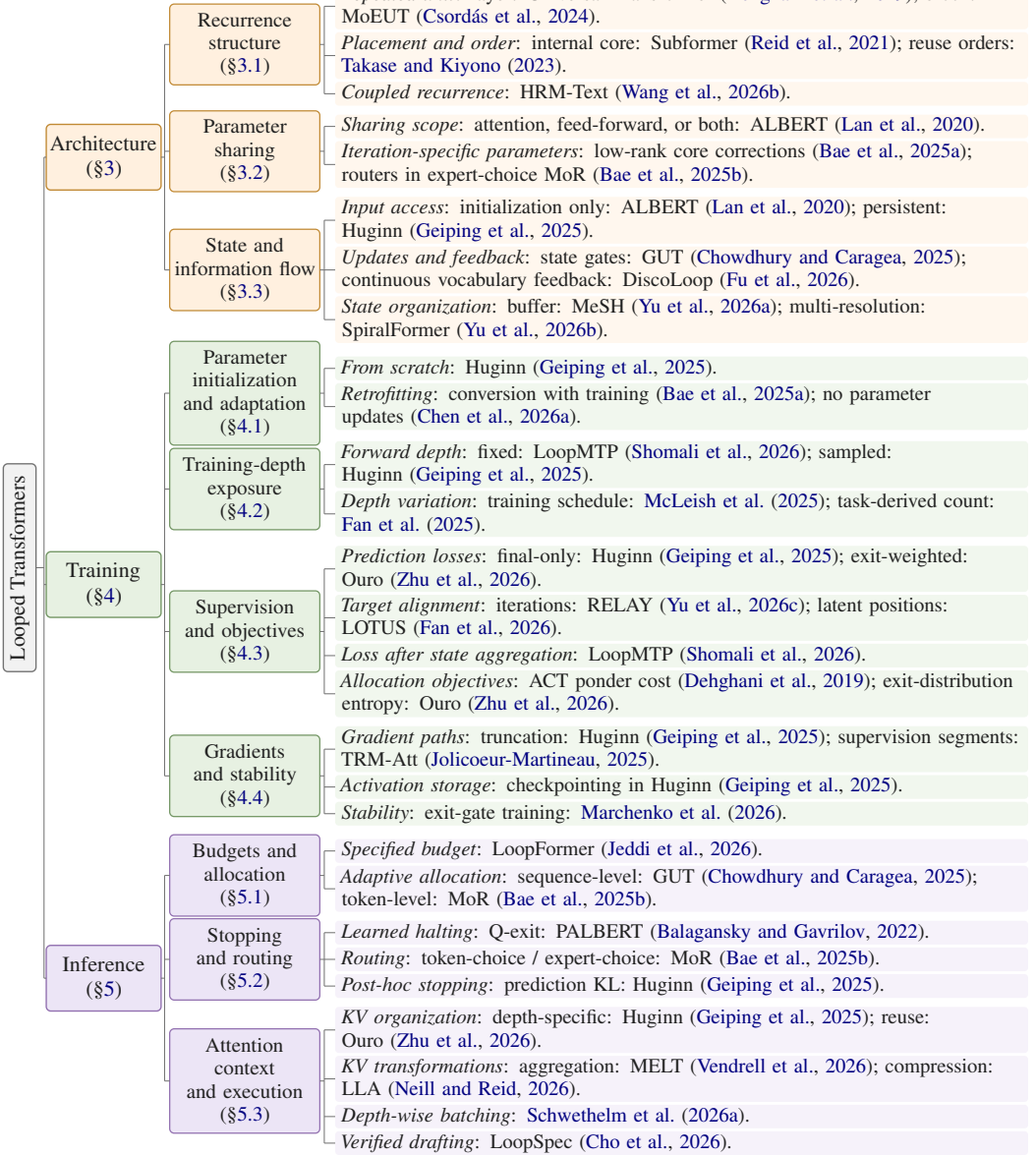

Looped Transformers
Architecture (§3)
Recurrence structure (§3.1)
Repeated unit: layer: Universal Transformer (Dehghani et al., 2019); block: MoEUT (Csordás et al., 2024).
Placement and order: internal core: Subformer (Reid et al., 2021); reuse orders: Takase and Kiyono (2023).
Coupled recurrence: HRM-Text (Wang et al., 2026b).
Parameter sharing (§3.2)
Sharing scope: attention, feed-forward, or both: ALBERT (Lan et al., 2020).
Iteration-specific parameters: low-rank core corrections (Bae et al., 2025a); routers in expert-choice MoR (Bae et al., 2025b).
State and information flow (§3.3)
Input access: initialization only: ALBERT (Lan et al., 2020); persistent: Huginn (Geiping et al., 2025).
Updates and feedback: state gates: GUT (Chowdhury and Caragea, 2025); continuous vocabulary feedback: DiscoLoop (Fu et al., 2026).
State organization: buffer: MeSH (Yu et al., 2026a); multi-resolution: SpiralFormer (Yu et al., 2026b).
Training (§4)
Parameter initialization and adaptation (§4.1)
From scratch: Huginn (Geiping et al., 2025).
Retrofitting: conversion with training (Bae et al., 2025a); no parameter updates (Chen et al., 2026a).
Training-depth exposure (§4.2)
Forward depth: fixed: LoopMTP (Shomali et al., 2026); sampled: Huginn (Geiping et al., 2025).
Depth variation: training schedule: McLeish et al. (2025); task-derived count: Fan et al. (2025).
Supervision and objectives (§4.3)
Prediction losses: final-only: Huginn (Geiping et al., 2025); exit-weighted: Ouro (Zhu et al., 2026).
Target alignment: iterations: RELAY (Yu et al., 2026c); latent positions: LOTUS (Fan et al., 2026).
Loss after state aggregation: LoopMTP (Shomali et al., 2026).
Allocation objectives: ACT ponder cost (Dehghani et al., 2019); exit-distribution entropy: Ouro (Zhu et al., 2026).
Gradients and stability (§4.4)
Gradient paths: truncation: Huginn (Geiping et al., 2025); supervision segments: TRM-Att (Jolicoeur-Martineau, 2025).
Activation storage: checkpointing in Huginn (Geiping et al., 2025).
Stability: exit-gate training: Marchenko et al. (2026).
Inference (§5)
Budgets and allocation (§5.1)
Specified budget: LoopFormer (Jeddi et al., 2026).
Adaptive allocation: sequence-level: GUT (Chowdhury and Caragea, 2025); token-level: MoR (Bae et al., 2025b).
Stopping and routing (§5.2)
Learned halting: Q-exit: PALBERT (Balagansky and Gavrilov, 2022).
Routing: token-choice / expert-choice: MoR (Bae et al., 2025b).
Post-hoc stopping: prediction KL: Huginn (Geiping et al., 2025).
Attention context and execution (§5.3)
KV organization: depth-specific: Huginn (Geiping et al., 2025); reuse: Ouro (Zhu et al., 2026).
KV transformations: aggregation: MELT (Vendrell et al., 2026); compression: LLA (Neill and Reid, 2026).
Depth-wise batching: Schwethelm et al. (2026a).
Verified drafting: LoopSpec (Cho et al., 2026).